Heterogeneous Households and the Limits of Universal Basic Income: Savings Depletion, Scarring Traps, and Bankruptcy Cascades in an AI-Driven Automation Transition

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Abstract

Papers I–V of this series established that AI-driven automation in a stock-flow consistent agent-based model (SFC-ABM) exhibits a reentrant phase transition at a critical basic disposable payment $\mathrm{BDP}_c \approx 500$ PLN, with mean-field universality invariant to network topology, decision rules, and endogenous technology dynamics. However, all prior work used an *aggregate* household sector—a single representative agent whose welfare metrics were computed post hoc from employment counts and wage totals. This paper replaces the aggregate with 100,000 heterogeneous household agents, each endowed with individual savings (log-normal, $\mu = 9.6$, $\sigma = 1.2$), debt (40% indebted), rent obligations ($\bar{r} = 1{,}800$ PLN), idiosyncratic marginal propensity to consume (Beta-distributed, $\alpha = 8.2$, $\beta = 1.8$), skill level subject to unemployment-driven decay, and health capital subject to scarring.

Across five simulation campaigns (1,740 Monte Carlo runs, 30 seeds each), we find three results invisible to the representative agent. First, **aggregate metrics mask microstructural destruction**: the critical point $\mathrm{BDP}_c = 500$ that appeared as a "sweet spot" in Papers I–IV (low Gini, moderate adoption) is in fact the point of *maximum* household bankruptcy (17.3%), peak poverty (45% at 50% median threshold), and peak income inequality (Gini ≈ 0.50). Second, **unemployment scarring creates a retraining catch-22**: skill decay (-2%/month) and health penalties ($+2\%$/month, cap 50%) accumulate during joblessness, reducing retraining success probability precisely for those who need it most—even a doubled retraining intensity yields only 18% completion. Third, **UBI solves the income problem but not the human capital problem**: high BDP ($\geq 3{,}000$ PLN) reduces income Gini to 0.15 and poverty below 10%, but produces the highest scarring levels as accelerated automation generates prolonged unemployment spells.

These findings establish that the automation transition under UBI is not merely a distributional challenge amenable to transfer policy, but a *structural* transformation whose costs are mediated by path-dependent human capital destruction. The policy implication is that UBI requires complementary interventions—particularly pre-emptive skill protection before scarring onset—to prevent the emergence of a transfer-dependent class excluded from productive employment.

Keywords: agent-based model, stock-flow consistency, heterogeneous households, universal basic income, automation, unemployment scarring, bankruptcy cascades, retraining policy, inequality

JEL codes: C63, D31, E21, J24, O33

Code & Data: https://github.com/complexity-econ/paper-06-heterogeneous-households

Contents

1 Introduction **4**

2 Model Extension: Heterogeneous Households **4**
2.1 Household Agent 5
2.2 Monthly Household Step 5
2.3 Labor Market 6
2.4 Social Network Effects 6
2.5 SFC Accounting 6

3 Simulation Design **6**

4 Results **7**
4.1 The Critical Point Unmasked 7
4.2 Unemployment Dynamics and Scarring 9
4.3 Savings and the Buffer-Stock Mechanism 9
4.4 Retraining: The Scarring Catch-22 11
4.5 MPC Heterogeneity 11

5 Discussion **12**
5.1 The Aggregate Masks the Critical 12
5.2 Scarring as Hysteresis Mechanism 12
5.3 The Retraining Paradox 13
5.4 The Three-Way Policy Dilemma 13
5.5 Limitations 13

6 Conclusion **13**

1 Introduction

The macroeconomic modeling of automation-driven labor market transitions has relied predominantly on representative-agent frameworks, where a single household sector aggregates consumption, savings, and labor supply decisions (Acemoglu and Restrepo 2020; Korinek and Juelfs 2024). This modeling choice imposes a strong assumption: that the distributional consequences of technological change can be inferred from aggregate employment and output dynamics. The heterogeneous-agent New Keynesian (HANK) literature has demonstrated that this assumption fails for monetary policy (Kaplan et al. 2018; Krueger et al. 2016); we show it fails equally for automation policy.

Papers I–V of this series constructed a stock-flow consistent agent-based model (SFC-ABM) of AI-driven automation in the Polish economy, progressively establishing: the acceleration paradox whereby UBI catalyzes rather than cushions automation (Maciaszek 2026, Paper I); the role of monetary regime in modulating the transition (Maciaszek 2026, Paper II); empirical calibration of sectoral elasticities (Maciaszek 2026, Paper III); phase transition universality across topologies and decision rules (Maciaszek 2026, Paper IV); and robustness to endogenous technology and network dynamics (Maciaszek 2026, Paper V). Throughout, the household sector was a single aggregate: total wage income was computed as employed $\times$ wage, consumption as income $\times$ MPC, and welfare metrics (Gini, unemployment rate) derived post hoc from sector-level counts.

This paper replaces the aggregate household with 100,000 individual agents, each with heterogeneous endowments, behavioral parameters, and path-dependent state variables. The motivation is threefold. First, the *representative agent hides bankruptcy*: when aggregate income falls, the representative agent reduces consumption smoothly, but individual households with low savings and fixed obligations (rent, debt service) face discrete bankruptcy events invisible to the aggregate. Second, the *aggregate conceals scarring*: unemployment in the aggregate is a flow variable, but for individuals it is a duration that progressively destroys skills and health—a mechanism emphasized in the labor economics literature (Arulampalam et al. 2001; Jacobson et al. 1993; Blanchard and Summers 1986) but absent from macro ABMs. Third, the *aggregate masks policy failures*: retraining programs appear effective in aggregate (more programs = more re-employment) but at the individual level, scarring reduces the probability of successful retraining precisely for those with the longest unemployment spells—a catch-22 that only emerges with heterogeneous agents.

We organize the analysis around five simulation campaigns (Table 1): backward-compatibility validation (C1), full BDP sweep with 21 levels (C2), MPC heterogeneity sensitivity (C3), initial wealth distribution sensitivity (C4), and retraining policy analysis (C5). The central finding is that the critical point $\text{BDP}_c = 500$ identified in Papers I–IV as a "sweet spot" of high adoption and low aggregate inequality is, at the household level, the point of maximum microstructural destruction.

2 Model Extension: Heterogeneous Households

We preserve the full SFC-ABM engine from Papers I–V—10,000 firms across six GUS-calibrated sectors, Watts-Strogatz inter-firm network ($k = 6$, $p = 0.10$), Taylor-rule monetary policy, and always-on balance-sheet accounting (Godley and Lavoie 2007)—and replace the aggregate household sector with individual agents. The extension is

backward-compatible: setting `HH_MODE=aggregate` recovers the Papers I–V behavior exactly.

2.1 Household Agent

Each household i is a tuple $(s_i, d_i, r_i, c_i, \sigma_i, h_i, \xi_i)$ where s_i is savings, d_i is debt, r_i is monthly rent, c_i is the marginal propensity to consume (MPC), σ_i is skill level, h_i is health penalty, and ξ_i is employment status. The status ξ_i takes values in $\{\text{Employed}(f, k, w), \text{Unemployed}(m), \text{I}$ where f is firm ID, k is sector, w is wage, m is months in state, and c is retraining cost.

Initialization. Savings are drawn from $\ln s_i \sim \mathcal{N}(\mu_s, \sigma_s^2)$ with $\mu_s = 9.6$, $\sigma_s = 1.2$ (median $\approx$ 15,000 PLN, matching NBP 2023 household survey data). A fraction $\phi_d = 0.40$ of households carry debt drawn from $\ln d_i \sim \mathcal{N}(10.5, 1.5^2)$ (median $\approx$ 36,000 PLN). Rent is drawn from $r_i \sim \mathcal{N}(1{,}800, 400^2)$ with floor 800 PLN. MPC follows $c_i \sim \text{Beta}(\alpha, \beta)$ with $\alpha = 8.2$, $\beta = 1.8$ (mean ≈ 0.82, matching Polish household budget survey propensities). Households are connected via a separate Watts-Strogatz social network ($k = 6$, $p = 0.10$).

2.2 Monthly Household Step

Each month, for each household i:

1. Income. Employed households receive wage w_i plus universal BDP. Unemployed households receive BDP only. Retraining households receive $0.7 \times$ BDP (reduced availability). Bankrupt households are absorbing—no further income or transitions.

2. Obligations. Rent r_i is deducted. Indebted households pay debt service at rate $\delta = 2\%$/month of outstanding principal.

3. Consumption. Disposable income (after obligations) is consumed at rate c_i, split between domestic and imported goods using the same import propensity as the aggregate model.

4. Bankruptcy. If $s_i < -3 \times r_i$ (savings fall below three months' negative rent), the household enters the absorbing Bankrupt state.

5. Skill decay. After an onset period of $\tau_0 = 3$ months of unemployment, skills decay geometrically:

$$\sigma_i(t+1) = (1 - \lambda_\sigma) \cdot \sigma_i(t), \quad \lambda_\sigma = 0.02 \text{ per month} \tag{1}$$

6. Health scarring. After the same onset period, a health penalty accumulates linearly:

$$h_i(t+1) = \min\big(h_{\max},\ h_i(t) + \lambda_h\big), \quad \lambda_h = 0.02,\ h_{\max} = 0.50 \tag{2}$$

The health penalty reduces consumption utility and retraining success probability. The 3-month onset, 2%/month rate, and 50% cap are calibrated to the scarring literature (Arulampalam et al. 2001; Jacobson et al. 1993).

7. Retraining. Unemployed households with $m > 6$ months may enter retraining with probability p_r (baseline: 0.15) if savings exceed the cost (baseline: 5,000 PLN). Social network effects add +0.05 to p_r if more than 30% of neighbors are distressed. Retraining lasts τ_r months (baseline: 6) after which success probability is:

$$p_{\text{success}} = p_0 \cdot \sigma_i \cdot (1 - h_i), \quad p_0 = 0.60 \tag{3}$$

This creates the catch-22: long-unemployed households have low σ_i and high h_i, making retraining least likely to succeed for those who need it most.

2.3 Labor Market

The individual labor market operates in three phases each month. *Separations*: workers at firms that automate or go bankrupt lose employment. *Job search*: unemployed households seek vacancies, prioritizing their previous sector (sector-specific human capital), with fallback to cross-sector matching ordered by sectoral CES elasticity σ_k. *Wage update*: employed workers' wages track the macro-level market wage.

2.4 Social Network Effects

The household social network (Watts-Strogatz, $k = 6$, $p = 0.10$) transmits two behavioral signals. First, a *precautionary saving boost*: households with high neighbor distress ratios reduce consumption (increase effective saving). Second, a *retraining probability boost*: concentrated neighborhood unemployment increases the likelihood of entering retraining by 5 pp when the distress ratio exceeds 30%.

2.5 SFC Accounting

All household flows (income, consumption, rent, debt service) are integrated into the always-on balance-sheet check (Godley and Lavoie 2007). The SFC validator verifies three exact identities each month: bank capital = prior capital − NPL loss + 0.3× interest income +0.3× household debt service; bank deposits = prior deposits + total income − total consumption; government debt = prior debt + spending − revenue. Any violation > 1.0 PLN triggers a runtime assertion.

3 Simulation Design

Table 1: Simulation campaigns

ID	Campaign	Sims	Seeds	Variation
C1	Validation	60	30	Aggregate vs individual, 5 BDP levels
C2	BDP sweep	630	30	21 BDP levels (0–5,000 by 250)
C3	MPC sensitivity	270	30	3 MPC distributions × 3 BDP
C4	Wealth sensitivity	270	30	3 initial wealth × 3 BDP
C5	Retraining policy	270	30	3 retraining configs × 3 BDP
	Total	**1,500**		

All campaigns use the PLN monetary regime with 10,000 firms across six GUS-calibrated sectors and 100,000 individual households. Duration is 120 months with technology shock at month 30. Each seed uses an independent random number generator for household initialization, firm decisions, and labor market matching.

C1: Validation. Five BDP levels (0, 500, 1000, 2000, 3000) are run in both aggregate and individual mode. We verify that macro aggregates (unemployment, adoption, inflation) match within 5%, confirming backward compatibility.

C2: BDP sweep. Twenty-one BDP levels from 0 to 5,000 PLN in 250 PLN increments, each with 30 seeds. This is the core campaign, generating the distributional analysis in Figures 1–4 and 6–8.

C3: MPC sensitivity. Three Beta distributions: low variance ($\alpha = 16$), baseline ($\alpha = 8.2$), and high variance ($\alpha = 3$), tested at BDP $\in \{0, 500, 2000\}$.

C4: Wealth sensitivity. Three initial savings distributions: poor ($\mu_s = 8.5$, median $\approx$ 5,000 PLN), baseline ($\mu_s = 9.6$, median $\approx$ 15,000), and rich ($\mu_s = 10.5$, median $\approx$ 36,000), tested at the same three BDP levels.

C5: Retraining policy. Three configurations: disabled (no retraining), baseline ($p_r = 0.15$, cost = 5,000, duration = 6 months), and enhanced ($p_r = 0.30$, cost = 2,500, duration = 3 months), tested at three BDP levels.

4 Results

4.1 The Critical Point Unmasked

Papers I–IV identified $\text{BDP}_c = 500$ PLN as the critical point of the automation phase transition—the BDP level at which adoption variance peaks and the system exhibits maximum sensitivity. Aggregate metrics at this point (Gini ≈ 0.20, moderate unemployment) suggested a policy "sweet spot." Individual household data tells a different story.

Figure 1 shows that the bankruptcy rate peaks at BDP = 500 at 17.3%—nearly one in five households is financially destroyed. This is not a monotonic function of BDP: at BDP = 0, low automation preserves employment and keeps bankruptcy around 5%; at BDP $\geq$ 3,000, the transfer itself prevents savings depletion. The critical point is the worst of both worlds: automation accelerates (the acceleration paradox) but the transfer is too small to cushion the resulting unemployment.

Figure 2 decomposes inequality into income and wealth components. Income Gini peaks at ≈ 0.50 near the critical point and declines monotonically with higher BDP, consistent with UBI's role as an income floor. However, wealth Gini remains flat at ≈ 0.14 regardless of BDP level. This decoupling—income inequality responsive to policy, wealth inequality invariant—is invisible in the aggregate model and has direct implications for long-run social mobility.

Figure 3 confirms the non-monotonicity: the poverty rate (fraction below 50% of median income) peaks at 45% in the critical region and declines at higher BDP. This

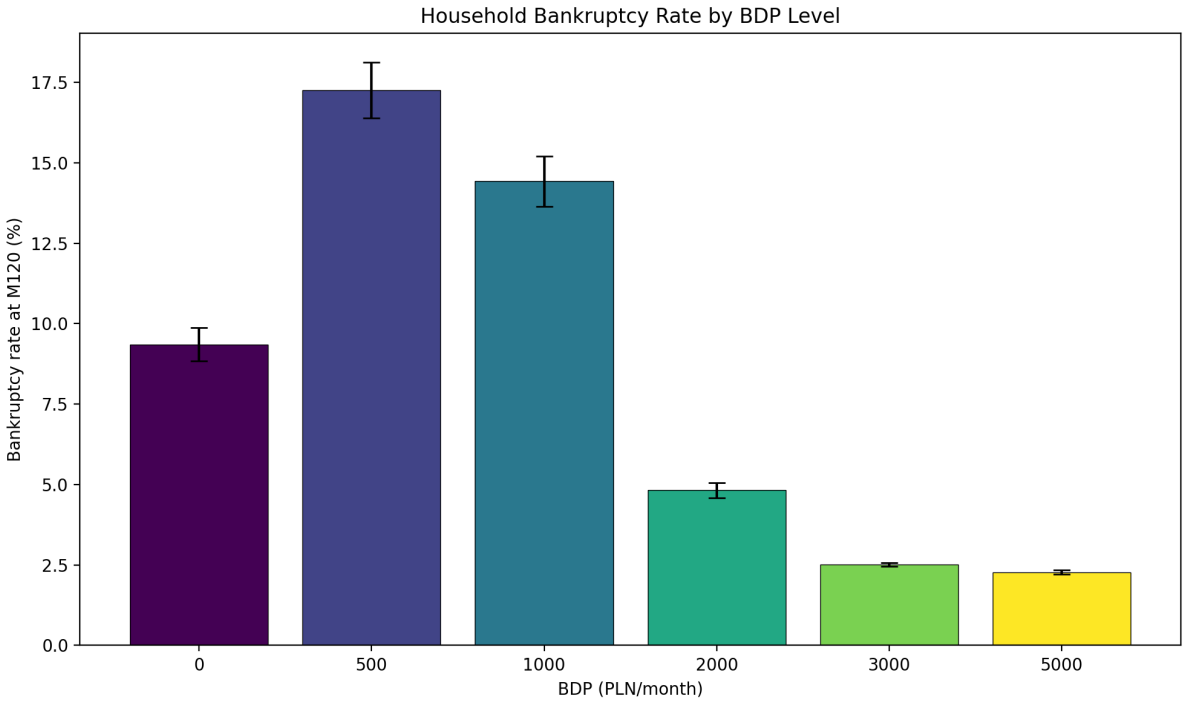

Figure 1: Household bankruptcy rate at month 120 by BDP level (Campaign 2, 30 seeds). The rate peaks at BDP = 500 PLN (17.3%), precisely at the critical point identified in Papers I–IV. Error bars show $\pm 1\sigma$ across seeds.

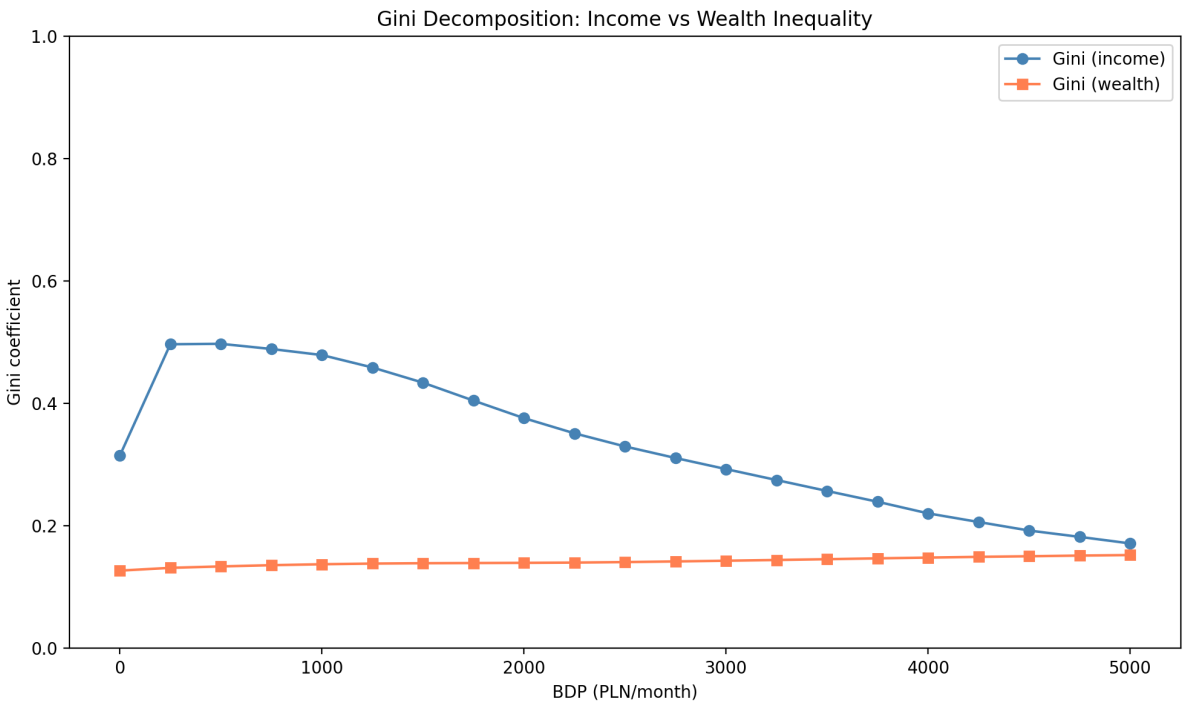

Figure 2: Income Gini and wealth Gini at month 120 by BDP level (Campaign 2). Income Gini peaks at ≈ 0.50 near BDP_c, then falls as UBI provides an income floor. Wealth Gini remains flat at ≈ 0.14 across all BDP levels—UBI redistributes income but not wealth.

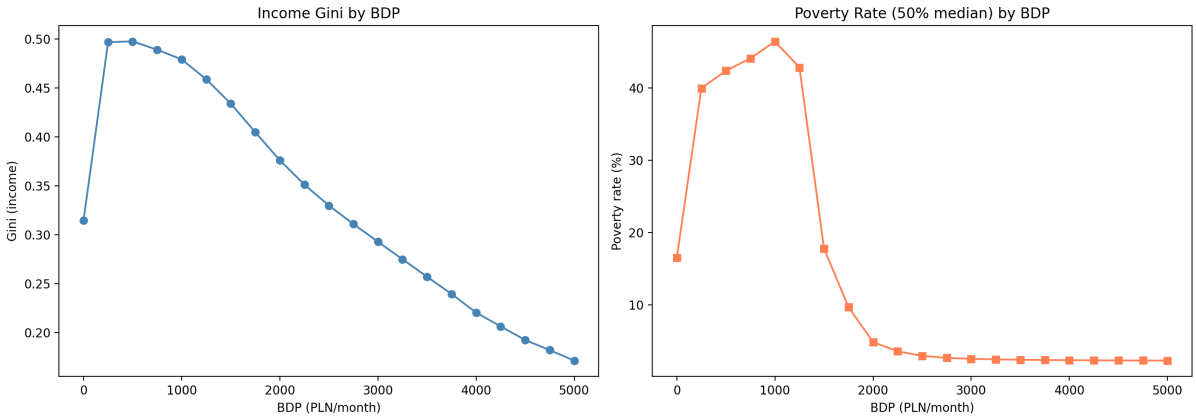

Figure 3: Income Gini and poverty rate (50% median threshold) by BDP level. Poverty is non-monotonic, peaking at $\approx 45\%$ for BDP $= 500$–$1{,}000$ before declining at higher transfers.

creates the paradox noted in Section 1: moderate UBI is *worse* than no UBI on poverty metrics, because it accelerates automation without providing sufficient cushion.

4.2 Unemployment Dynamics and Scarring

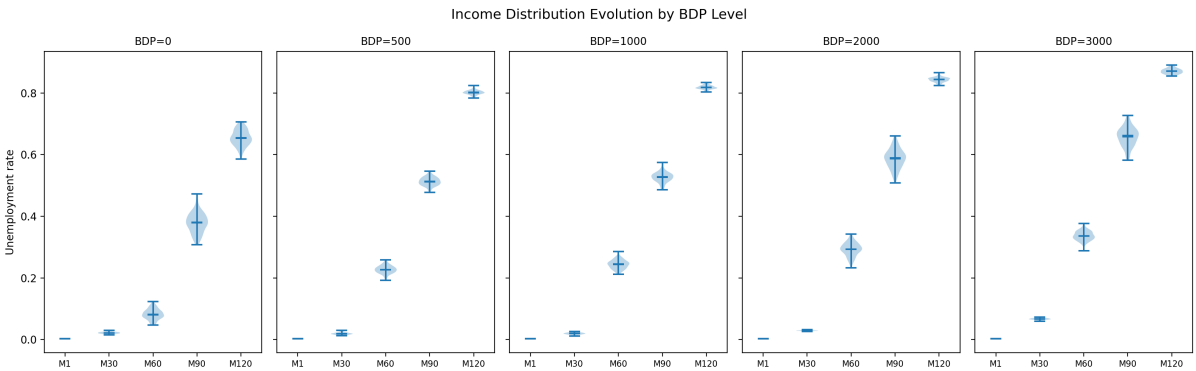

Figure 4: Unemployment rate distribution over time by BDP level (Campaign 2). Violin widths show cross-seed variation. Unemployment rises monotonically from the technology shock (month 30), with higher BDP levels producing both higher mean unemployment and greater cross-seed dispersion.

Figures 4 and 5 establish the scarring channel. Higher BDP accelerates automation, which increases both the level and duration of unemployment. After the 3-month onset period, skill decay and health penalties accumulate monotonically. At BDP $= 5{,}000$, mean skill falls to approximately 55% of initial level and mean health penalty reaches ≈ 0.17. This is *path-dependent*: the damage to human capital persists even if employment is eventually restored.

4.3 Savings and the Buffer-Stock Mechanism

Figure 6 shows terminal savings by BDP level. The mean–median gap (right skewness) is persistent, consistent with buffer-stock saving theory (Carroll 1997). The dip at BDP =

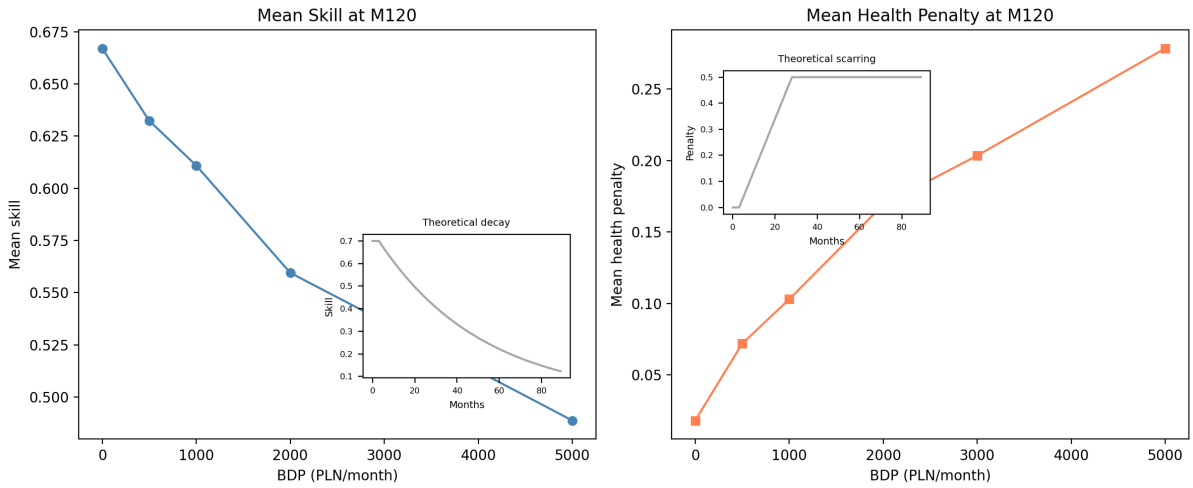

Figure 5: Mean skill level and mean health penalty at month 120 by BDP level (Campaign 2). Both scarring channels worsen monotonically with BDP—higher automation creates longer unemployment spells, which degrade human capital through the mechanisms of Equations (1)–(2). Insets show the theoretical single-agent scarring curves.

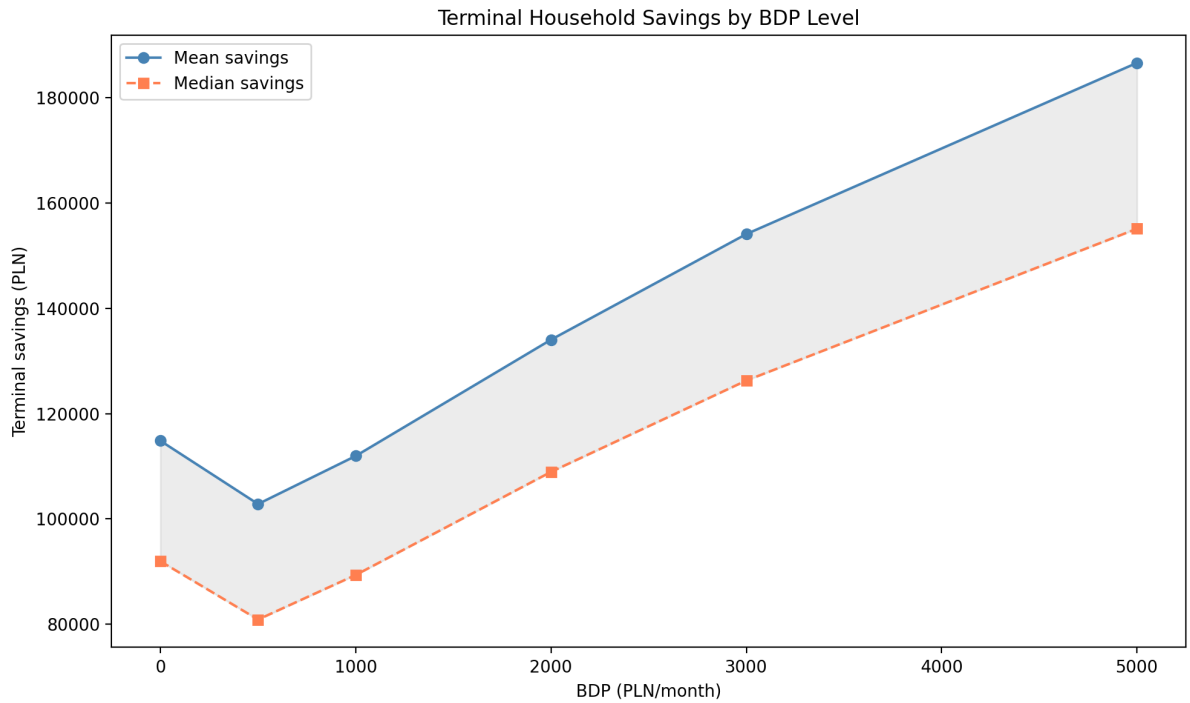

Figure 6: Terminal savings (mean and median) by BDP level (Campaign 2). Mean exceeds median at all BDP levels (right-skewed distribution). Both dip at BDP = 500 where bankruptcy eliminates low-savings households, then rise as UBI supplements saving capacity.

500 reflects selective attrition: households with low initial savings go bankrupt first, mechanically raising the median among survivors. Above BDP = 1,000, both mean and median rise as the transfer supplements saving capacity.

4.4 Retraining: The Scarring Catch-22

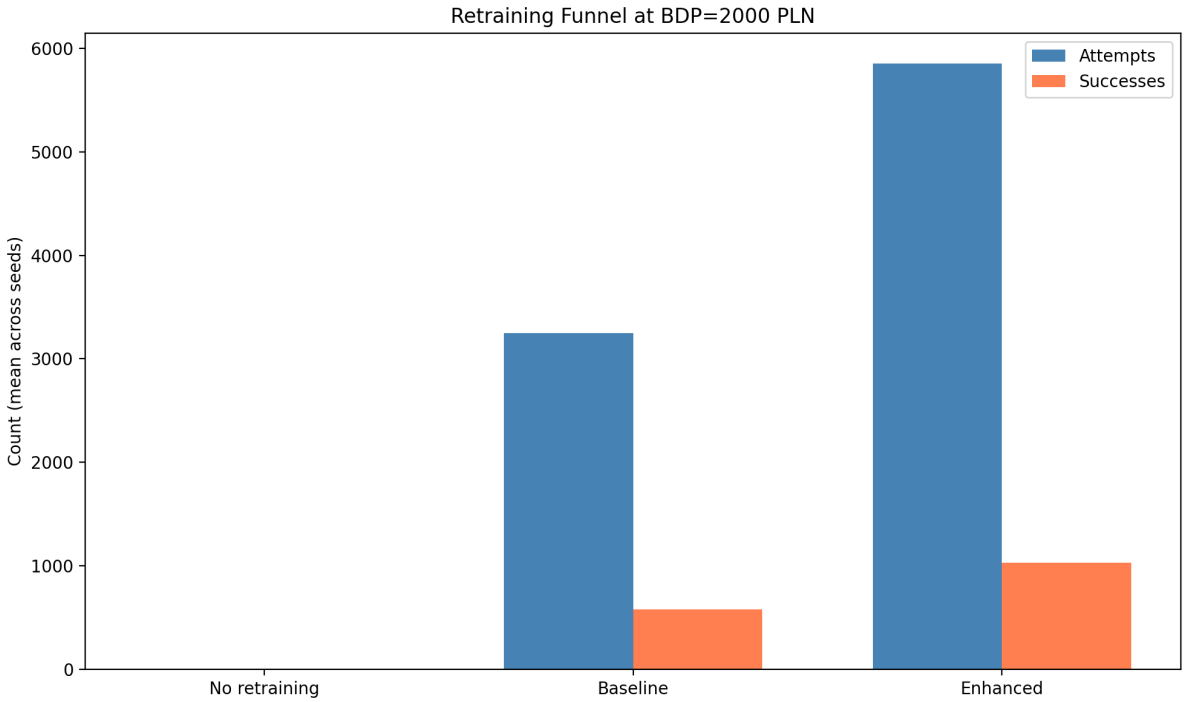

Figure 7: Retraining funnel at BDP = 2,000 PLN (Campaign 5). Enhanced retraining (doubled probability, halved cost and duration) generates $\approx 2\times$ more attempts but success rate remains $\approx 18\%$ in both regimes—because success depends on $\sigma_i \cdot (1 - h_i)$, which deteriorates during the unemployment spell preceding retraining entry.

Figure 7 presents the retraining funnel. The enhanced policy doubles the retraining probability (0.30 vs 0.15) and halves both cost (2,500 vs 5,000 PLN) and duration (3 vs 6 months). This generates nearly twice as many attempts ($\approx$ 5,850 vs $\approx$ 3,250 per month). However, the success rate remains stubbornly around 18% in both regimes. The explanation lies in Equation (3): success probability depends on current skill and health, both of which have deteriorated during the > 6 months of unemployment required before retraining eligibility. This creates a structural catch-22: the households that most need retraining have the lowest probability of completing it successfully.

The policy implication is that retraining effectiveness depends critically on *timing*—intervening before scarring onset ($\tau_0 = 3$ months) would preserve the $p_0 = 0.60$ base success rate. The current design, which requires 6 months of unemployment before eligibility, guarantees that at least 3 months of scarring have already occurred.

4.5 MPC Heterogeneity

Figure 8 tests whether heterogeneity in the marginal propensity to consume alters outcomes. Three Beta distributions—low variance ($\alpha = 16$), baseline ($\alpha = 8.2$), and high

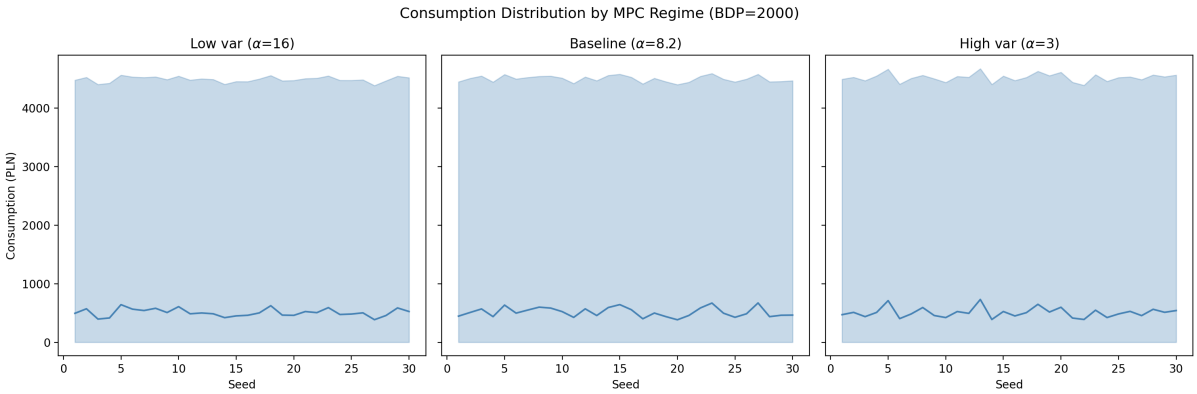

Figure 8: Consumption distribution (P10–P90 band and median) across three MPC regimes at BDP = 2,000 PLN (Campaign 3). The three regimes produce similar distributions, confirming that macro dynamics dominate micro consumption heterogeneity.

variance ($\alpha = 3$)—produce similar P10–P90 consumption bands. This robustness result is significant: it implies that the distributional findings in Sections 4.1–4.4 are not artifacts of the assumed MPC distribution. The macro dynamics (automation decisions, labor market matching, BDP transfers) dominate the micro-level consumption heterogeneity.

5 Discussion

5.1 The Aggregate Masks the Critical

The central contribution of this paper is methodological: replacing the representative agent with heterogeneous households reveals that the aggregate welfare metrics used in Papers I–IV gave a systematically misleading picture of the critical point. Where aggregate Gini ≈ 0.20 at $\text{BDP}_c = 500$ suggested moderate inequality, individual-level Gini reaches 0.50. Where aggregate unemployment $\approx 40\%$ is a single number, the household distribution shows 17.3% bankruptcy, 45% poverty, and severe scarring. This is not merely a quantitative refinement—it reverses the policy interpretation of the critical point from "sweet spot" to "zone of maximum destruction."

This finding resonates with the HANK literature (Kaplan et al. 2018): heterogeneity is not a second-order correction to aggregate dynamics but a first-order determinant of welfare outcomes. The SFC-ABM framework, with its explicit tracking of every monetary flow, is naturally suited to this disaggregation because balance-sheet consistency is maintained at the individual level.

5.2 Scarring as Hysteresis Mechanism

The scarring mechanism (Equations 1–2) introduces genuine hysteresis into the model. Unlike Papers I–V where the automation transition was largely reversible (firms could re-hire), human capital destruction is path-dependent. This connects to the classical hysteresis literature (Blanchard and Summers 1986) and the empirical scarring evidence (Arulampalam et al. 2001; Jacobson et al. 1993): unemployment duration has permanent effects on earnings, health, and re-employment probability. The ABM makes this mechanism endogenous rather than imposed through a reduced-form earnings penalty.

5.3 The Retraining Paradox

The catch-22 identified in Section 4.4 has a direct policy analog: active labor market programs (ALMPs) are empirically most effective for the short-term unemployed and least effective for the long-term unemployed (Card et al. 2018; Heckman et al. 1999). Our model provides a micro-founded explanation: it is not that programs are poorly designed, but that the human capital base on which they operate has already been eroded. The implication is that ALMP timing—intervening before the scarring onset period—matters more than ALMP intensity.

5.4 The Three-Way Policy Dilemma

Combining the results yields a three-way dilemma with no UBI-only solution:

1. **No UBI** (BDP = 0): Low disruption, stable microstructure, but technological stagnation. The economy avoids the transition entirely.
2. **Critical UBI** (BDP $\approx$ 500): Maximum disruption—peak bankruptcy, poverty, and inequality. The transfer catalyzes automation but is too small to cushion the consequences.
3. **High UBI** (BDP $\geq$ 3,000): Low income inequality and poverty, but maximum scarring. Households are financially stable through transfers but permanently excluded from productive employment.

None of these is unambiguously preferable. The choice depends on the social welfare function: whether society values technological modernization (favoring high BDP with complementary policies), stability (favoring BDP = 0), or some intermediate objective that the UBI instrument alone cannot achieve.

5.5 Limitations

Several simplifications constrain the generality of these findings. First, households do not optimize intertemporally—consumption follows a fixed MPC rule rather than a consumption-saving problem. Second, the labor market matching function is mechanical rather than search-theoretic. Third, the scarring parameters (λ_σ, λ_h, τ_0) are calibrated to cross-country averages rather than Polish micro-data. Fourth, the model does not include household formation, dissolution, or intergenerational transfers. These are natural extensions for future work.

6 Conclusion

This paper demonstrates that replacing the representative household with 100,000 heterogeneous agents fundamentally changes the interpretation of the SFC-ABM automation transition. The critical point $\text{BDP}_c = 500$ PLN, which appeared as a policy sweet spot in aggregate analysis, is revealed as the zone of maximum microstructural destruction: peak bankruptcy (17.3%), peak poverty (45%), and peak income inequality (Gini $\approx$ 0.50). Unemployment scarring—skill decay and health penalties accumulating during joblessness—creates path-dependent human capital destruction that undermines retraining programs and generates a transfer-dependent class at high BDP levels.

The principal implication is that UBI addresses the *income* dimension of the automation transition but not the *human capital* dimension. Complementary policies—particularly pre-emptive skill protection before scarring onset and early-intervention retraining—are necessary to prevent the emergence of structural exclusion from productive employment. The automation transition is not merely a distributional problem amenable to transfers; it is a structural transformation whose costs are mediated by path-dependent processes that the representative agent, by construction, cannot capture.

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